# Mark R. Sepanski


### Contact Information

Baylor University
Department of Mathematics
One Bear Place #97328
Waco, TX 76798-7328 USA

+1 (254) 366-9200
MRSepanski@gmail.com
Mark_Sepanski@baylor.edu
www.baylor.edu/math/index.php?id=54018


### Interests

Mathematics, Lie groups, representation theory, combinatorics, applications of mathematics, research, mathematics education, program development, growth, strategy, creative vision, and management

### Education

1990 to 1994: **Massachusetts Institute of Technology** Cambridge, MA

Ph.D. in Mathematics, May 1994
Thesis title: *$L_2(q)$ and the rank two Lie groups: their construction, geometry, character formulas, and invariant theory in light of Kostant's conjecture*
Thesis advisor: Professor Bertram Kostant

1987 to 1990: **Purdue University** West Lafayette, IN

B.S. in Mathematics, *With Highest Distinction*, May 1990
Special project under Professor Steve Bell: *Numerical methods in conformal mapping and the Szegö kernel function*

### Employment

2010 to present: **Graduate Program Director** Waco, TX
Baylor University Mathematics
recruitment, managament, program development, training, strategy, online education

1997 to present: **Baylor University** Waco, TX
Current Position: Professor

1995 to 1997: **Oklahoma State University** Stillwater, OK
Position: Visiting Assistant Professor

1994 to 1995: **Cornell University** Ithaca, NY
Position: Visiting Assistant Professor

1993 to 1994: **Massachusetts Institute of Technology** Cambridge, MA
Position: Teaching Assistant

1990: **Cummins Diesel Engine Corporation** Columbus, IN
Position: Summer Intern in Quality Management and Statistics Department.
Duties included design of experiments, ANOVA with the Taguchi method, and employee training in these techniques

### Publications

1. Partitions with designted summands, joint with D. Herden, et. al., *in final preparation.*
2. Counting the parts divisible by $k$ in all partitions of $n$ whose parts have multiplicity less than $k$, joint with D. Herden, et. al., *submitted.*
3. A case study of full lecturer-provided notes for teaching university mathematics in a compressed format, *submitted.*
4. Explicit Pieri Inclusions, joint with M. Hunziker and J. Miller, *submitted*
5. Schrödinger-type equations and unitary highest weight representations of $U(n,n)$, joint with M. Huziker and R. Stanke, *J. Lie Theory.*, 30 (2020), no. 1, 201–222.

6. Lyndon word decompositions and pseudo orbits on $q$-nary graphs, joint with R. Band and J. Harrison, *J. Math. Anal. Appl.*, 470 (2019), no. 1, 135–144.
7. Schrödinger-type equations and unitary highest weight representations of the metaplectic group, joint with M. Hunziker and R. Stanke, Representation theory and harmonic analysis on symmetric spaces, 157–174, *Contemp. Math.*, 714, Amer. Math. Soc., Providence, RI, 2018.
8. Net regular signed trees, joint with I. Michael, *Australas. J. Combin.* 66 (2016), 192–204.
9. Factorizations of relative extremal projectors, joint with C. Conley, *p-Adic Numbers Ultrametric Anal. Appl.* 7 (2015), no. 4, 276–290.
10. A System of Schrödinger Equations and the Oscillator Representation, joint with M. Hunziker and R. Stanke, *Electron. J. Differential Equations*, Vol. 2015 (2015), No. 260, pp. 1-28.
11. Global representations of the conformal group and eigenspaces of the Yamabe operator on $S^1 \times S^n$, joint with J. Franco, *Pacific J. Math.* 275 (2015), no. 2, 463–480.
12. On divisibility of convolutions of central binomial coefficients, *Electron. J. Combin.*, 21, no. 1, Paper 1.32, 7 pp, 2014.
13. Global Lie symmetries of a system of Schrödinger equations and the oscillator representation, joint with M. Hunziker and R. Stanke, *Miskolc Math. Notes*, Vol. 14, No. 2, pp. 647–657, 2013.
14. Global representations of the heat and Schrödinger equation with singular potential, joint with J. Franco, *Electron. J. Differential Equations*, 2013(154), 1-16, 2013.
15. Polynomials and Weyl algebras on time scales, joint with M. Hunziker, *Rocky Mountain J. Math*, 42(6), 1893-1918, 2012.
16. The minimal representation of the conformal group and classical solutions to the wave equation, joint with M. Hunziker and R. Stanke, *J. Lie Theory*, 22, 301-360, 2012.
17. Global Lie symmetries of the heat and Schrödinger equation, joint with R. Stanke, *J. Lie Theory* 20, no. 3, 543–580, 2010.
18. The nonlinear potential filtration equation and global actions of Lie symmetries, *Electron. J. Differential Equations*, No. 101, 24 pp, 2009.
19. Distinguished orbits and the L-S category of simply connected compact Lie groups, joint with M. Hunziker, *Topology Appl.* 156, 2443-2451, 2009.
20. Lie symmetries for the nonlinear heat equation, *J. Math. Anal. Appl.* 360, 35-46, 2009.
21. Positivity of zeta distributions and small unitary representations, joint with L. Barchini and R. Zierau, The ubiquitous heat kernel, *Contemp. Math.*, 398:1-46, Amer. Math. Soc., Providence, RI, 2006.
22. Infinite commutative product formulas for relative extremal projectors, joint with C. Conley, *Adv. Math.*, 196:52-77, 2005.
23. On global $SL(2,\mathbb{R})$ symmetries of differential operators, joint with R. Stanke, *J. Funct. Anal.*, 224:1-21, 2005.
24. Singular projective bases and the projective Bol operator, joint with C. Conley, *Adv. in Appl. Math.*, 33(1):158-191, 2004.
25. Relative extremal projectors, joint with C. Conley, *Adv. Math.*, 174:155-166, 2003.

26. Schur orthogonality relations for the finitary infinite symmetric group, joint with R. Donley, *Linear and Multilinear Algebra*, 50(4):373-377, 2002.
27. Finite-reductive dual pairs in $G_2$, joint with L. Barchini, *Linear Algebra Appl.*, 340:123-136, 2002.
28. Application of restriction of Fourier transforms to an example from representation theory, joint with L. Barchini, *Pacific J. Math.*, 198(2):265-293, 2001.
29. K-types of $SU(1,n)$ representations and restriction of cohomology, *Pacific J. Math.*, 192(2):385-398, 2000.
30. Block-compatible metaplectic cocycles, joint with W. Banks and J. Levy, *J. Reine Angew. Math.*, 507:131-163, 1999.
31. Closure ordering and the Kostant–Sekiguchi correspondence, joint with D. Barbasch, *Proc. Amer. Math. Soc.*, 126(1):311-317, January 1998.
32. Boundaries of $K$-types, restriction of cohomology, and the multiplicity free case, *Compositio Math.*, 106(1):31-41, 1997.
33. Kostant's conjecture and $L_2(q)$ invariant theory in the rank two Lie groups, *Comm. Algebra*, 24(6):1915–1938, 1996.
34. Kostant's conjecture and the geometry of Cartan subalgebras in the rank two and exceptional Lie groups, *J. Algebra*, 182:347–382, 1996.
35. $L_2(q)$ and the rank two Lie groups: their construction in light of Kostant's conjecture, *Trans. Amer. Math. Soc.*, 347(10):3983–4021, October 1995.

## Books

1. *Algebra*, Pure and Applied Undergraduate Texts, vol. 11, American Mathematical Society, 2010.
2. *Compact Lie Groups*, Graduate Texts in Mathematics, Springer-Verlag, 2006.

## Honors

2019 to present: **Mathematics Advisory Group** (MAG) for Graduate Education for Transforming Post-Secondary Education in Mathematics (TPSEMath)

2018 to present: **Mentor** for the Math Alliance (ensures that every underrepresented student with the talent and the ambition has the opportunity to earn a doctoral degree in a mathematical science)

2014 to present: **Editor** European Journal of Mathematics

2001 Fall: Member at **Mathematical Sciences Research Institute**, Berkeley, CA.

## Awards Grants

2021: **NSA Grant**, Southwest Local Algebra Meeting 2021, Principal Investigator: *awarded.*

2002 to 2003: **NSA Young Investigator Grant**, MDA 904-02-1-0081, Principal Investigator: *Relative extremal projectors and Lie algebra cohomology.*

1996 to 1999: **NSF Research Grant**, DMS 9623280, Principal Investigator: *Boundaries of K-types and restriction of cohomology.*

1990 to 1993: **NSF Graduate Research Fellowship**.

1990: **Michael Golom Prize for the Outstanding Undergraduate Math Student**, Purdue University.

1988 to 1990: **Barry M. Goldwater National Scholarship**.

1988 to 1990: **Mercury Seven Foundation National Science Scholarship**.

**PhD Students**

John Miller (joint with M. Hunziker)
Jose Franco

**Teaching Experience**

Pre-Calculus, Calculus 1, Calculus 2, Calculus 3, Linear Algebra, Ordinary Differential Equations, Partial Differential Equations, Introduction to Analysis, Advanced Calculus 1, Advanced Calculus 2, Theory of Functions of Real Variables 1, Theory of Functions of Real Variables 2, Functional Analysis, Complex Analysis, Advanced Complex Analysis, Foundations of Combinatorics and Algebra, Abstract Algebra, Advanced Abstract Algebra 1, Advanced Abstract Algebra 2, Topology, Algebraic Topology 1, Algebraic Topology 2, Differential Geometry, Lie Algebras, Compact Lie Groups, Lie Groups, Semisimple Lie Groups, Advanced Topics in Representation Theory, Crystal Bases